

FAST-WEIGHT MATRIX STATES ARE A REPRESENTATIONAL MEDIUM, NOT AN EFFICIENCY TRICK

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ABSTRACT

Fast-weight and linear-attention models maintain a matrix-valued state, written by an outer-product update and evaluated mainly as an efficiency substitute for attention. We ask instead what the matrix state stores, at what dimensionality, and what the storage costs at scale: three pre-registered findings. First, a rank law: in a matrix-state encoder family trained on five permutation-group word problems (minimal faithful representation dimension 2 to 5), recruited effective rank tracks $d_{\min}$ (Spearman $\rho = 0.9747$, the tie-capped maximum), a designed S_4 -versus- A_5 test shows rank follows dimension, not solvability, and forcing rank one below $d_{\min}$ zeroes recovery while $d_{\min}$ restores it. Second, a capability separation: a two-layer delta-rule model performs episodic recall at accuracy 0.9995 against a parameter-matched vector-state ablation at chance, the pre-registered decisive comparison; a compute-matched transformer also fails at or below chance after an explicit four-point learning-rate search. The capability resides causally in the first layer’s state, is linearly legible only after downstream nonlinear processing, and holds at 0.998 or higher to 1798 tokens on a fixed 32,768-byte state while that transformer, cache-capped, reads chance everywhere. Third, the same write mechanism drives a population-geometry pathology that worsens monotonically from 14M to 1.31B parameters. Removing qk-normalization leaves it unchanged, and a frozen-key-bias mitigation fails to remove it: only the mitigation’s loss neutrality survives the move to scale.

1 INTRODUCTION

Linear-attention and fast-weight sequence models replace the growing key-value cache of a transformer with a fixed-size matrix state, written by an outer-product rule and read by a matrix-vector product. The delta-rule family (DeltaNet and its gated descendants; Yang et al., 2025b;a) is the current standard bearer: at each step the state is updated as $S_t = S_{t-1}(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$, a rank-one correction that overwrites the value previously associated with the incoming key. The literature evaluates these models almost exclusively on one axis: quality per unit of compute or memory relative to attention. The state itself is treated as an implementation detail.

We take the opposite view: the $d \times d$ state itself is the object of study, a representational medium with a capacity structure and a causal role of its own, and with failure modes that never appear on a perplexity leaderboard. This paper measures all three, under pre-registered verdict criteria, across three experimental programs that share one object of study: what the matrix state stores.

Our thesis, scoped per leg to the architecture family that carries it: matrix-valued state representations are a genuine representational medium. In a matrix-state encoder family trained on group word problems, stochastic gradient descent recruits exactly the task-minimal state rank, and forcing one rank fewer destroys the task while restoring that rank restores it. In a two-layer delta-rule model, the first layer’s matrix state causally carries an episodic-recall capability that a parameter-matched vector-state ablation lacks at equal training budget (a compute-matched transformer also fails, after an explicit learning-rate search), and the stored content is linearly legible only after downstream nonlinear processing. The same write mechanism, run at language-model scale, drives a population-geometry pathology that worsens monotonically across a two-decade parameter ladder and survives the community’s stock mitigation.

Concretely, we contribute:

1. **A rank law, causally closed** (Section 3). Across five permutation groups whose minimal faithful representation dimensions $d_{\min}$ span 2 to 5, recruited effective rank tracks $d_{\min}$ at Spearman $\rho = 0.9747$, the maximum the family’s tie structure permits, with 19 of 19 unconstrained cells inside the pre-registered band. A designed pair, S_4 versus A_5 , shares $d_{\min} = 3$ but differs in solvability; an equivalence test declares their recruited ranks equivalent; solvability leaves no signature. Train-time rank forcing closes the causal loop: recovery is 0.000 at $k = d_{\min} - 1$ in all five groups and in all four seeds of the extension group, and returns at $k = d_{\min}$ in five of five.
2. **A capability separation at matched budget** (Section 4). On single-pass episodic recall, the delta-rule contender reads accuracy 0.99951 against 0.03394 for its vector-state ablation; the paired confidence interval excludes the pre-registered 0.30 margin, its floor sitting at more than three times the margin, and this ablation comparison carries the verdict. A compute-matched transformer also reads chance (0.02832 at the shared default); a subsequent four-point learning-rate search on the recall task itself (10^{-4} to 3×10^{-3} , three seeds, 20,000 matched steps) finds no rate that clears the bar, and the best-optimizing rate reads recall furthest below chance, dissociating optimization quality from recall. The transformer is accordingly a second failing baseline, not a second verdict (Sections 4.1, 6, and 7). Zeroing the first layer’s state collapses recall to chance while zeroing the second layer’s changes nothing, and no state-level linear probe reads the content that the model’s own forward pass decodes. The capability is also memory-stable: accuracy 0.998 or higher out to 1798 tokens on a fixed 32,768-byte state, while a transformer whose cache is capped at 1 to 32 times that budget reads chance everywhere.
3. **A pathology that scale makes worse and the stock fix does not remove** (Section 5). The write keys of delta-rule language models drift toward a collapsed population geometry; the span fraction of that drift climbs 0.248 to 0.455 across a 14M-to-1.31B ladder at held-fixed data mixes. Removing qk normalization changes nothing (0.05σ at $n=3$), so the attractor is not an artifact of the standard stabilizer, and a frozen-key-bias mitigation that is loss-neutral at every scale fails to transfer its 14M geometric benefit to 98M or 392M.
4. **A mechanism scaffold** (Appendix A). In the encoder family the trained state decomposes as $Z \approx c^*I +$ a rank- $(K-1)$ task correction at 0.5 to 2.9 percent residual, and deviation from this scaffold is a loss-blind health signal; the corresponding channel is empty by construction in delta-rule states, which bounds how far the scaffold generalizes.

Sections 6 and 7 carry the capability result of Section 4 onto two further axes, binding breadth and parameter scale, inside a 98M-to-392M language-model graft.

Every numerical claim in this paper traces to a pre-registered verdict record and an archived, check-summed raw artifact; the figures regenerate from those artifacts under checksum assertion. Each repaired instrument reading appears together with the defect it corrected; Section 2 summarizes this discipline and Section 7 discusses its scope limits.

2 BACKGROUND AND EXPERIMENTAL SETUP

2.1 DELTA-RULE FAST-WEIGHT STATES

A delta-rule layer maintains a matrix state $S \in \mathbb{R}^{d \times d}$ updated per token as $S_t = S_{t-1}(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$, with keys k_t , values v_t , and write strength β_t produced from the token stream; reads are matrix-vector products $S_t q_t$. The update first erases the value currently bound to k_t , then writes v_t at k_t : an associative store with in-context writes. Because outer-product writes confine S to the span of the observed keys and values, the state’s effective dimensionality and the geometry of the key population are both measurable objects, and both are studied here.

2.2 MODEL FAMILIES

Three model families appear, one per result leg. (i) The *matrix-state encoder* family (Section 3): a transformer encoder that emits a single $d_{\text{state}} \times d_{\text{state}}$ matrix Z per input word under a hard single-

state bottleneck; the decoder reads only Z , never the raw inputs, so the task load must reside within the one matrix. (ii) The *two-layer delta-rule contender* (Section 4): a DeltaNet-family language model with $d_{model} = 256$, two blocks, $d_{state} = 64$, and its two matched baselines, described in Section 4.1. (iii) *Delta-rule language models at scale* (Section 5): the same architecture family trained as causal LMs at 14M, 98M, 392M, and 1.31B parameters on held-fixed data mixes.

2.3 TASKS

Group word problems (Section 3). A word $w = g_{i_1} \cdots g_{i_L}$ is drawn over a symmetric generating set of a permutation group G ; the target is the image of the composed element under a fixed faithful orthogonal representation. Five groups are used: S_3, S_4, A_5, S_5, A_6 , with minimal faithful real representation dimensions $d_{min} = 2, 3, 3, 4, 5$. The pair S_4/A_5 is the designed dissociation control: equal d_{min} , opposite solvability. Train words have length $L \leq 8$; held-out evaluation words are fresh draws.

Episodic recall (Section 4). Each episode binds $K=32$ key-entity/value-entity pairs (real GPT-2 token identities, injectivity-guarded) over a 224-token bind phase, then queries one key. The metric of record, acc_A , is episode-restricted 32-way top-1 accuracy at the answer position, read through each model’s own LM-head route; chance is $1/32 = 0.03125$ and the pre-registered demonstration bar is three times chance, 0.09375. Every acc_A number in this paper carries the same caveat: recall here means episode-restricted top-1 retrieval under argmax decoding, and under argmax a rank-one state can support on the order of d associations (Nichani et al., 2024); no rank or continuous-capacity claim is made from acc_A .

Language modeling at scale (Section 5). Causal LMs are trained on two fixed data mixes (a reasoning mix built on OpenR1-Math and a narrative mix built on WikiText-103, each with a fixed augmentation source), held identical across the parameter ladder so that scale is the only moving axis.

2.4 INSTRUMENTS

Restricted effective rank (Section 3). States $Z(w)$ for held-out words are centered, the dominant d_{min} -dimensional subspace U is taken from the SVD of the centered second-moment matrix, and effective rank (the entropy-based Roy-Vetterli measure) is computed on $U^\top Z(w)U$; fitting and evaluation use disjoint word splits. Centering is load-bearing: without it, an ambient identity block masquerades as signal.

Force-rank arms (Section 3). At train time, Z is projected to rank k by SVD truncation inside the forward pass, for $k \in \{d_{min} - 1, d_{min}, d_{min} + 1\}$; recovery is measured as the fraction of held-out words whose Procrustes-aligned cosine to the target exceeds 0.9 (xrec90).

State zeroing and probe taps (Section 4). Causal attribution zeroes one layer’s cached state before the query pass and re-measures acc_A ; representational legibility is measured by ridge probes at four taps (three state-level, one post-block pre-LM-head), reported as the recovered fraction at cosine 0.9 (rf@0.9) against a shuffled control.

Span fraction (Section 5). For a population of K normalized write keys E , the raw instrument is the key-Gram deviation $\|E^\top E - I\|_F$; span fraction rescales it between two analytic anchors, the expectation for K i.i.d. random unit vectors (value 0) and full collapse onto one direction (value 1), making readings comparable across state widths. Higher is closer to collapse.

Confidence intervals. Paired confidence intervals in this paper are Student- t intervals on $n=3$ per-seed paired differences (two degrees of freedom, $t_{.975} = 4.303$), except the nine-seed diagnosis pool of Section 4.5, stated there. Per-seed values are tabulated in Appendix B so any alternative construction can be applied; the verdicts that rest on these intervals clear their margins by multiples of the interval width, and the one threshold-adjacent reading claimed as a trend (Section 5.2) is reported as unconfirmed.

2.5 INSTRUMENT VALIDITY AS A FIRST-CLASS METHOD

Each leg of this program survived at least one instrument-defect-found-and-fixed round; only repaired readings appear below. The rank program’s first causal sweep was voided by a target-padding defect that let rank-capped models buy cosine from an ambient identity block; the diagnosis, fix, and re-run are described in Section 3.3. The recall program lost three consecutive probe rounds to a wrong-layer, wrong-route linear instrument before the repaired tap of Section 4.3 read what the model’s own forward pass had decoded all along. A third case came from a separate composition program (not claimed here): a primary lens read zero on converged models while a pre-registered crosscheck lens read one, the contradiction was resolved by a recorded tiebreak against the raw artifacts, and the affected endpoint is excluded from this paper entirely. Every verdict-bearing instrument here therefore carries an analytic anchor or a negative control, with a pre-registered crosscheck wherever a verdict hangs on a single metric.

3 THE RANK LAW

In the encoder family, the task’s representation theory sets the state’s dimensionality.

3.1 RECRUITED RANK TRACKS MINIMAL FAITHFUL DIMENSION

Figure 1 shows recruited restricted effective rank against $d_{\min}$ for the five groups, per seed. The per-group means are 1.877 (S_3), 2.852 (S_4), 2.832 (A_5), 3.591 (S_5), and 4.736 (A_6) against $d_{\min} = 2, 3, 3, 4, 5$. Spearman correlation across the family is $\rho = 0.9747$. This number is the maximum the family permits: the S_4/A_5 tie at $d_{\min} = 3$ caps the achievable midrank-adjusted ρ at exactly 0.9747, and the observed ordering attains that cap. Because the family has only five members, we report the exact permutation null rather than an asymptotic one: under the null, $P(\rho \geq 0.8) = 8/120 \approx 6.67\%$. The correlation is therefore suggestive on its own; the band criterion and the causal test below do the deciding. The tracking also holds cell by cell: all 19 unconstrained-arm cells in the sweep land inside the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band.

3.2 THE DESIGNED DISSOCIATION: S_4 VERSUS A_5

S_4 is solvable; A_5 is the smallest non-solvable group. Both have $d_{\min} = 3$. If recruited rank were driven by computational difficulty in the Krohn-Rhodes sense, the pair should separate; if by representation dimension, the pair should land together. We pre-registered a two-one-sided-tests equivalence procedure (Welch, $n = 5$ per group, margin ± 0.5 rank-units, half the ladder’s adjacent-rung spacing). The result is a declaration of equivalence: mean difference 0.0194 rank-units (2.852 versus 2.832), standard error 0.0368, $df = 7.83$, with both one-sided statistics ($t_1 = 13.06$, $t_2 = 14.12$) exceeding the critical value 1.865 by a factor of seven. Recruited rank tracks the representation dimension of the task; the group’s algebraic complexity class contributes nothing detectable.

3.3 AN INSTRUMENT DEFECT, ITS DIAGNOSIS, AND THE REPAIRED CAUSAL TEST

An early force-rank sweep produced a mechanically valid FALSIFY reading that a target-embedding defect later voided, and the defect changes what the causal test means. The training target had been embedded as $\rho(g) \oplus I_{d_{\text{state}} - d_{\min}}$: an eye-padded block matrix of full rank d_{state} with all singular values equal to one. For such a target, a rank- k model’s optimal direct cosine is $\sqrt{k/d_{\text{state}}}$ independent of which subspace it spends its rank on, and the harvest matched this prediction: 36 of 39 force-rank cells sat within 0.07 of the predicted ceiling, one further cell sat marginally outside at 0.072, and the two remaining outliers (0.15 and 0.17, both in one group’s below- $d_{\min}$ arm) reflect an additional optimization shortfall. Rank-capped models were buying cosine from the identity padding, so the causal boundary at $d_{\min}$ was never tested. The fix replaced the padding with zeros, making the target rank exactly $d_{\min}$, and the decisional metric was pre-registered before the re-run as the Procrustes-aligned crosscheck (xrec90), because the scale-only primary metric reads cosine 0.01 to 0.22 on a flawless oracle when the target’s rho-block eigenvalues are degenerate.

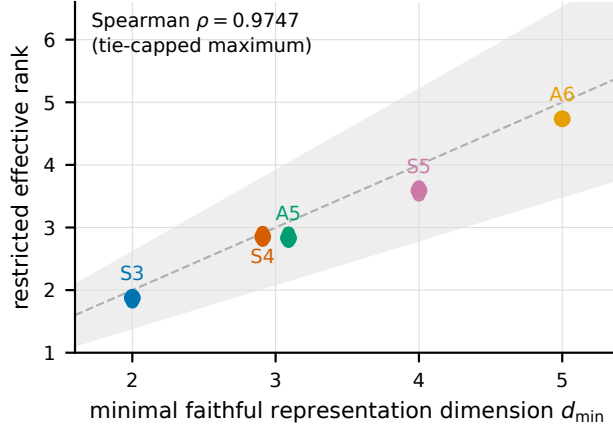


Figure 1: Recruited restricted effective rank versus minimal faithful representation dimension $d_{\min}$ for five permutation groups (S_3, S_4, A_5, S_5, A_6 ; unconstrained arms, per-seed points, group means marked). The shaded region is the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band; all 19 cells land inside it. Spearman $\rho = 0.9747$ equals the tie-capped maximum for this family. The S_4/A_5 pair (both $d_{\min} = 3$, opposite solvability) lands together; the equivalence test in Section 3.2 declares them equivalent within ± 0.5 rank-units.

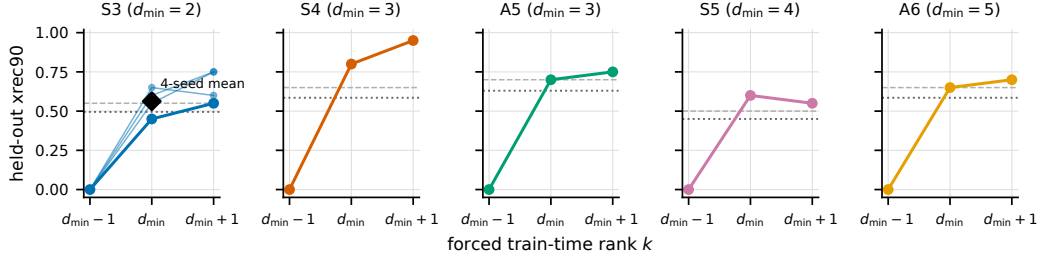


Figure 2: Held-out recovery (xrec90, Procrustes-aligned cosine threshold 0.9) versus forced train-time rank k , per group, on the zero-padded (repaired) target. Recovery is 0.000 at $k = d_{\min} - 1$ in every group and returns at $k = d_{\min}$; dashed lines mark each group's bar of 0.9 times its own unconstrained anchor. The S_3 panel overlays all four extension seeds: every seed reads 0.000 at $k = d_{\min} - 1$, and the seed-mean at $k = d_{\min}$ (0.5625) clears the bar (0.495).

3.4 RECOVERY IS A STEP FUNCTION AT EXACTLY $d_{\min}$

Figure 2 shows the repaired force-rank staircase. At $k = d_{\min} - 1$, held-out recovery xrec90 is 0.000 in all five groups: not degraded, zero. At $k = d_{\min}$, recovery returns and clears the pre-registered bar of 0.9 times each group's own unconstrained anchor in four of five groups directly (S_4 0.800 against bar 0.585; A_5 0.700 against 0.630; S_5 0.600 against 0.450; A_6 0.650 against 0.585). S_3 landed within the pre-stated marginality window (0.450 against bar 0.495) and was routed to its pre-registered seed extension: across four independent seeds the seed-mean at $k = d_{\min}$ is 0.5625, clearing the bar, and $k = d_{\min} - 1$ reads exactly 0.000 in all four seeds. The pre-registered confirmation criterion (a razor step at $d_{\min}$ in at least four of five groups, including at least one of the marquee pair) is met with both marquee groups confirming.

Together the three readings close the loop in both directions: $d_{\min}$ is recruited when available and is causally necessary and sufficient when forced. The substrate is small and every reading reproduces from the archived artifacts (Appendix B).

Table 1: Episodic recall acc_A per seed and arm at matched parameters, tokens, and compute; chance $1/32 = 0.03125$, demonstration bar 0.09375.

arm	seed 0	seed 1	seed 2	mean
contender	0.99951	1.00000	0.99902	0.99951
vector-state ablation	0.03223	0.03271	0.03687	0.03394
transformer	0.02710	0.02930	0.02856	0.02832

4 CAPABILITY SEPARATION

The head-to-head comparison reported here was designed, margined, and frozen before its confirmatory sweep ran: a two-layer delta-rule model against a parameter-matched vector-state ablation and a compute-matched transformer, on single-pass episodic recall.

4.1 ARMS, MATCHING, AND FROZEN MARGINS

The contender is a two-block DeltaNet-family LM ($d_{\text{model}} = 256$, $d_{\text{state}} = 64$, 14,049,408 parameters). The ablation keeps the embedding table, output head, and feed-forward blocks identical and replaces only the fast-weight mixer with an elementwise-gated vector recurrence at the same d_{state} (14,048,384 parameters, a 0.007 percent difference); it is constructed as a recurrence rather than a reshape, which makes matrix structure the manipulated variable at near-identical parameter count. The transformer baseline is a two-layer pre-norm decoder with rotary position embeddings and the contender’s own MLP class, FLOP-matched to the contender within 5 percent. On this task its learning rate was the shared default 3×10^{-4} , identical to the contender’s and the ablation’s; the calibration record’s three-point learning-rate grid was run on the language-modeling control task, and no architecture-specific learning-rate search was performed for the transformer on the recall task. Its recall-task training loss is near flat across the run (7.81 to 7.84 at step 500, 7.45 to 7.51 at step 20,000, across the three seeds), a signature that raised the possibility of an under-optimized arm; Section 6 states the resulting interpretive limit and Section 7 reports the learning-rate search that resolves it. All arms train 20,000 steps at matched tokens with seeds identical across arms per cell. Verdict tiers, margins, and the primary task were frozen and checksummed before the confirmatory sweep launched (a WIN requires the paired per-seed difference’s 95 percent confidence interval to exclude 0.30); the sweep’s evaluation applied the audited round protocol verbatim to its 18 checkpoints with checksum-pinned loading and a fixed evaluation episode set.

4.2 THE VERDICT AT $n=3$

Table 1 (Figure 3, left) gives the per-seed metric of record.

Every contender seed exceeds the demonstration bar by a factor of at least 10.7; neither baseline clears it in any seed. The paired contender-minus-ablation difference is 0.96558 with 95 percent confidence interval (0.95822, 0.97293), and the contender-minus-transformer difference is 0.97119 with interval (0.96855, 0.97383); both intervals exclude the frozen 0.30 margin, the first by more than three times the margin at its floor. The pre-registered extension trigger (any seed below bar, or any interval straddling the margin) did not fire.

The Nichani caveat of Section 2.3 is part of the claim: recall means episode-restricted top-1 retrieval under argmax decoding, under which a rank-one state can support on the order of d associations, so nothing here is a rank or continuous-capacity claim. So is the matched-budget caveat: the transformer read holds only at this matched parameter count, token budget, and training compute. Because the transformer sits below its own demonstration bar at every point of the Section 7 learning-rate grid, it is recorded as a second failing baseline, and the separation verdict is carried by the ablation comparison with the transformer’s non-competitiveness disclosed alongside.

4.3 THE CAPABILITY IS FAST-WEIGHT-RESIDENT AND NONLINEARLY STORED

Causal attribution by state zeroing localizes the capability. On the frozen round checkpoint, zeroing the first block’s state before the query collapses contender recall from 0.9990 to 0.0286, at chance,

while zeroing the second block’s state leaves it at 0.9990, unchanged. The sweep replicates this on all fresh seeds: first-state zeroing reads 0.0339, 0.0012, and 0.0002 against the 0.09375 bar, and the hard-stop criterion is clean in 12 of 12 recurrent cells. The binding lives in S_0 ; S_1 is causally inert for this task.

Where the content becomes readable is a separate question. Ridge probes at three state-level taps, including directly on the causally load-bearing S_0 , recover nothing at the strict threshold ($\text{rf}@0.9 = 0.0$ at every state-level tap; the accompanying shuffled-control gaps of at most 0.063 are mean-cosine gaps, a separate continuous sanity check on a different scale from $\text{rf}@0.9$). The same probe at the post-block-1, pre-LM-head hidden state reads $\text{rf}@0.9 = 0.674$ with mean cosine 0.894. The ablation’s pre-LM-head tap, its own positive control, fails ($\text{rf}@0.9 = 0.0$, cosine 0.119); its pre-LM-head geometry is different in kind. The stored binding is therefore real but linearly illegible at the state; it becomes linearly decodable only after the downstream block’s nonlinear processing. A state-level linear probe alone would have read this capability as absent, while the model’s own forward pass decodes it at 0.9995.

4.4 CONSTANT-MEMORY RECALL HORIZON

The contender’s two states occupy $2 \times 64 \times 64$ float32 entries, 32,768 bytes, independent of context length. Holding those bytes fixed, recall does not decay with distance: at horizons of 2, 4, and 8 times the bind phase (454, 902, and 1798 tokens), every seed reads $\text{acc}_A \geq 0.998$. The pre-registered memory-matched comparison caps the transformer’s KV cache at M times the contender’s state bytes with sink-plus-FIFO eviction in the style of streaming attention caches (Xiao et al., 2024), $M \in \{1, 2, 4, 8, 16, 32\}$; every capped read at the decision horizon lies between 0.020 and 0.033, at or below chance, as does the uncapped read. Every per- M paired-gap confidence interval has a floor of at least 0.958 against the 0.20 crossover margin. Because the uncapped transformer itself fails the demonstration bar, the pre-registered degenerate-baseline clause fires: the verdict of record is that the baseline is non-competitive at matched parameters and tokens, not a strongest-possible-baseline result, and no crossover-point value is certified. The two informative readings are the contender’s own flat horizon table and the answer to the forced-locality question: capping never helps the transformer.

4.5 SCOPE DISCLOSURE: THE MULTI-HOP TASK

A second task in the frozen registry, two-hop compositional recall, is verdict-INDETERMINATE and is not claimed. Its data are disclosed because they constrain interpretation: one of three contender seeds clears the demonstration bar (0.33447; the other two and all baseline seeds read chance), the paired interval straddles the margin, the bar-clearing seed’s partial recall does not generalize to held-out hop depths (0.0112) and collapses to 0.010 at every extended horizon. The single bar-clearing seed suggested the failure mode is at least partly trainability variance rather than a hard capability boundary, and the pre-registered diagnosis round confirmed this: across nine pooled seeds, three contender seeds clear the demonstration bar (0.334, 0.391, 0.479) while zero of nine ablation seeds do; the hard-capability-boundary hypothesis is rejected for this task at this scale and budget. The pooled paired interval $(-0.020, +0.268)$ sits below the 0.30 margin and the strict-tier reading is a TIE, flagged non-decision-grade by a pre-registered batch-effect gate; every bar-clearing seed remains fragile at held-out hop depths and extended horizons. Nothing beyond single-hop recall is claimed here.

5 THE PATHOLOGY AT SCALE

The same write mechanism that Sections 3 and 4 show storing task structure also reshapes the population geometry of its own keys, for the worse, and more so at larger scale. The evidence here is independent of Sections 3 and 4.

5.1 THE WRITE-GEOMETRY ATTRACTOR WORSENS MONOTONICALLY WITH SCALE

In trained delta-rule LMs, the write-key population drifts away from the near-orthonormal geometry of random unit vectors toward a collapsed, low-dimensional configuration. Span fraction (Section

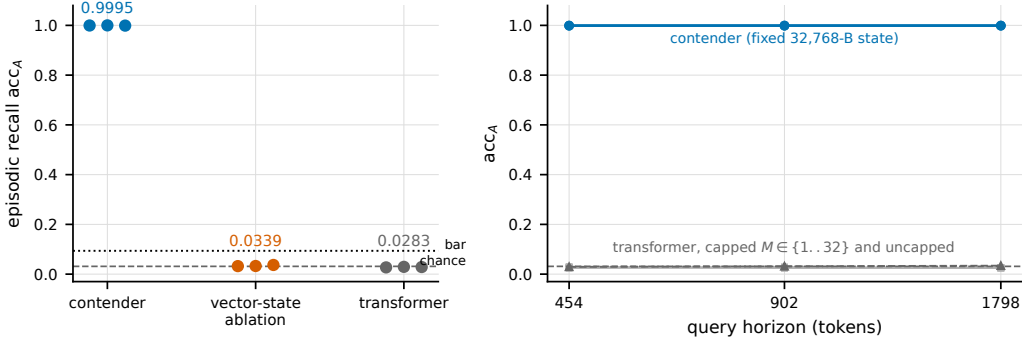


Figure 3: Left: episodic recall acc_A per seed and arm at matched parameters, tokens, and compute ($K=32$ bindings; chance 0.03125, dashed; demonstration bar 0.09375, dotted). The contender reads 0.99902 to 1.00000; both baselines sit at chance in every seed; paired intervals exclude the frozen 0.30 margin. Right: recall versus context horizon on a fixed 32,768-byte contender state (454 to 1798 tokens, all seeds at or above 0.998) against the KV-capped transformer grid ($M \in \{1..32\}$ times the contender’s state bytes, all reads 0.020 to 0.033). Recall means episode-restricted argmax retrieval (Nichani caveat, Section 2.3); the left-panel transformer is a second failing baseline after an explicit learning-rate search (Section 7), and the right-panel KV-capped transformer is a degenerate-baseline datum at this matched budget (Section 4.4).

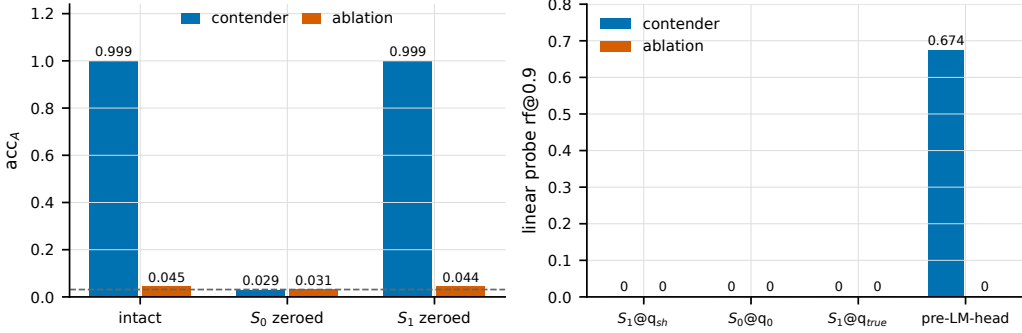


Figure 4: Storage localization and legibility for the contender and its vector-state ablation. Left: acc_A with both states intact, first state zeroed, and second state zeroed; first-state zeroing collapses the contender to chance (0.9990 to 0.0286) while second-state zeroing changes nothing. Right: linear-probe recovered fraction at cosine 0.9 for three state-level taps and the pre-LM-head tap; only the contender’s pre-LM-head tap reads above zero (0.674), so the binding stored in S_0 is linearly legible only after downstream nonlinear processing.

2.4) locates each model between the random anchor (0) and full collapse (1). Across a four-rung ladder trained on held-fixed data mixes, at 14M, 98M, 392M, and 1.31B parameters, pooled span fraction reads 0.248, 0.344, 0.389, and 0.455: monotone in scale, with the largest model halfway to collapse. Because the data mixes are held fixed across rungs, scale is the only moving axis.

The 1.31B cells self-terminated at their internal timeout at approximately 84.7 percent of the token-matched budget. The shortfall does not rescue the trend: over the final 25,000 recorded steps the rung’s span fraction is flat to mildly declining (0.4584 to 0.4554), so completing the budget would move the endpoint by roughly 0.003, more than twenty times smaller than the 0.066 increment from the previous rung.

5.2 REMOVING QK NORMALIZATION CHANGES NOTHING; GATING TRENDS TOWARD AMPLIFICATION

The entire ladder was trained with qk normalization active, the same in-kernel key normalization that production linear-attention stacks cite as their stability mitigation. A 2-by-2 factorial at 14M (qk-norm on/off crossed with delta-rule gating on/off, $n=3$ seeds, pre-registered same-corpus noise floor 2.244355 raw Gram-deviation units) gives both answers. Removing qk normalization changes the geometry reading by -0.103 , which is 0.05σ against the floor: a within-noise null. The attractor is measured with the mitigation active and is unchanged without it; it is not an artifact of the stabilizer. Turning on gating reads $+4.312$, which is 1.92σ , below the pre-registered 2σ confirmation bar of 4.489; all three paired seeds move in the amplifying direction and an exploratory Welch test gives $p = 0.062$. We record gating as a direction-consistent trend toward amplification: not a confirmed effect, and not a null.

5.3 THE SMALL-SCALE FIX DOES NOT TRANSFER

At 14M, a frozen-key-bias construction had moved the geometry: blending each key with a frozen per-token random unit anchor, $k \leftarrow \text{normalize}((1 - \lambda)k + \lambda B[\text{tok}])$ at $\lambda = 0.58$, with zero new trainable parameters. A pre-registered transfer wave tested it at 98M and 392M against an eval-retrofitted comparator (the baseline checkpoint measured through the identical blend, which removes the mechanical pinning artifact a naive comparison introduces), on both corpora, with a pre-blend co-primary instrument that measures the raw keys and is immune to the blend arithmetic by construction. The outcome is the pre-registered PARTIAL verdict at both scales. At 98M the deployed arm is destabilizing, the 14M sign, on both corpora and both instruments: span-fraction deltas $+0.1133$ with interval $(+0.0543, +0.1723)$ on the reasoning mix and $+0.1011$ with interval $(+0.0541, +0.1482)$ on the narrative mix. At 392M the narrative-mix delta is $+0.0189$ with interval $(+0.0112, +0.0266)$, still destabilizing, while the reasoning-mix interval includes zero. The effect reverses nowhere. A single-seed exploratory probe of the alternative global-vector construction decays from -0.33 and -0.23 at 14M to -0.058 and -0.034 at 98M, and at 392M reads -0.012 on one corpus and $+0.019$, a sign flip, on the other: the stabilization it bought at small scale decays toward zero and loses its sign. Meanwhile the constructions are free at the loss level: validation-loss neutrality passes its blind-pinned ceiling in all eight gate cells under the pre-registered arm-mean criterion; one individual seed in the 98M narrative-mix cell reads 3.2038 against the 3.2020 ceiling while its arm mean (3.1961) passes. The neutrality transfers to scale. The fix does not. The 392M token budget bounds the cross-scale comparison: those cells ran a reduced budget, so the apparent 98M-to-392M attenuation is confounded with tokens by design and is not a registered claim; the within-scale readings are the registered claims.

5.4 THE STORAGE MECHANISM AS COMMON FACTOR

The write that Sections 3 and 4 show storing task structure is the write that, iterated over a corpus at scale, drags its own key population toward collapse; the one construction known to stabilize the geometry at 14M buys nothing at 98M and 392M while costing nothing in loss. No fix is offered here: the community’s stock mitigation is orthogonal to the phenomenon, and the obvious portable construction fails to port. The pathology belongs to the storage mechanism as scaled; no stabilizer choice tested here explains it away, and it is currently unaddressed.

6 BREADTH SCALING

Section 4 measures a capability separation at one binding load. Two questions follow, and this paper now answers both. Does the separation survive more bindings, and does it survive more parameters? This chapter answers the first; Section 7 answers the second. The substrate changes: instead of the 14M two-block contender, the measurements below graft a small composition head onto a 98M-parameter delta-rule language model, so that what is being asked is whether a real language model can carry an in-context-written relation operator and compose it exactly.

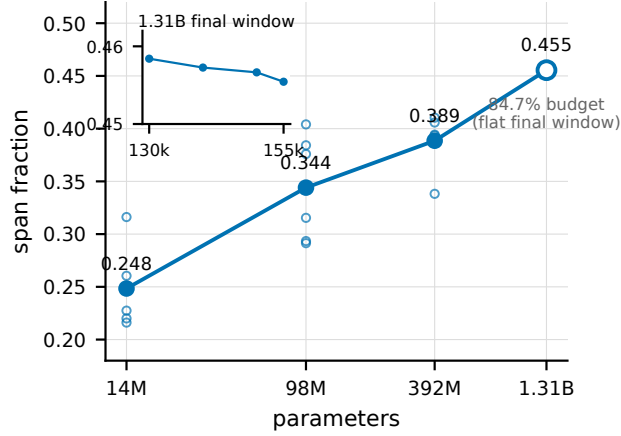


Figure 5: Write-key span fraction versus model scale (14M, 98M, 392M, 1.31B parameters; log-scale abscissa) on held-fixed data mixes; 0 is the random-unit-vector anchor and 1 is full collapse, computed per state width so rungs are comparable. The climb 0.248 to 0.455 is monotone. The 1.31B rung stopped at approximately 84.7 percent of its token budget with span fraction flat to declining over its final window (0.4584 to 0.4554); the missing budget cannot supply the 0.066-scale increment.

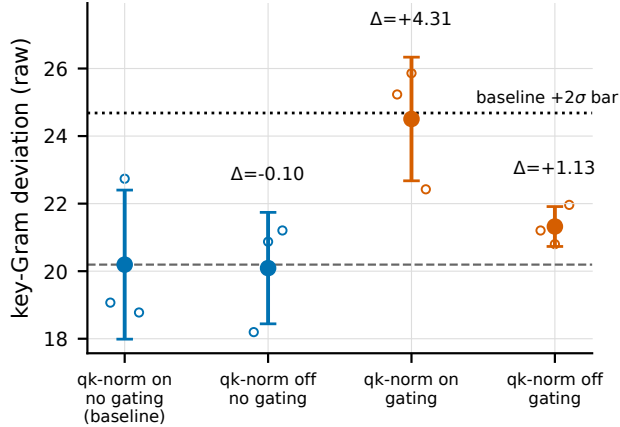


Figure 6: The 2-by-2 mitigation factorial at 14M ($n=3$; raw key-Gram deviation, cell means with per-seed points; dashed line: the pre-registered 2σ bar over the corrected same-corpus noise floor). Removing qk normalization is a within-noise null (-0.103 , 0.05σ): the attractor is not a qk-normalization artifact. Gating reads $+4.312$ (1.92σ , all three paired seeds positive), a direction-consistent trend below the confirmation bar: neither a confirmed amplification nor a null.

6.1 THE GRAFTED COMPOSITION HEAD AND ITS TWO WRITE REGIMES

The backbone is a DeltaNet-family causal language model (DeltaNet (Yang et al., 2025b) and Gated DeltaNet (Yang et al., 2025a)) with $d_{model} = 768$, 12 layers, $d_{state} = 64$, and 97,619,712 backbone parameters. Grafted onto it is a composition head that holds a single $d \times d$ relation operator O written from the context. K entities are arranged in one Hamiltonian K -cycle; the answer to a query at hop depth h is the image of the queried entity under O^h , which the read path computes by repeated squaring in $O(\log h)$ matrix multiplies rather than h sequential applications. The operator width is $d = K+1$ at every point, so the axis of this chapter is the *pair* (K, d) and never K alone.

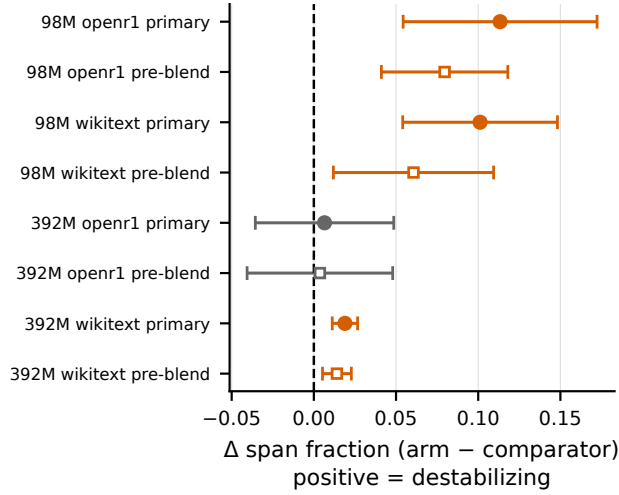


Figure 7: Transfer of the frozen-key-bias mitigation to scale: per-scale, per-corpus span-fraction deltas (deployed arm minus eval-retrofitted comparator, $n=3$, 95 percent intervals; negative would be stabilizing). At 98M both corpora read destabilizing with intervals excluding zero; at 392M one corpus reads destabilizing and one is null; no cell reverses. Validation-loss neutrality passes in all eight gate cells (not shown), so the construction’s cost neutrality transfers while its geometric benefit does not. The 392M rung ran a reduced token budget; cross-scale attenuation is therefore not a registered claim.

Two write regimes are measured at every point, through the identical read path and the identical trained weights.

- **P1b, exact teacher-forced operator substitution.** The operator is computed in closed form from the true binding and substituted into the trained checkpoint’s own read path. This is a read-path capability measurement and never a learned write; every capability number in this chapter and the next is P1b. - **P0, the model’s own SGD-learned write**, read through the identical path. This is the wall.

The metric is argmax retrieval over exactly the K episode slots at the answer position, reported chance-corrected as $\kappa = (\text{acc} - 1/K)/(1 - 1/K)$ so that readings at different K are comparable; chance is $1/K$. All curves of record use $n = 256$ evaluation episodes at eval seed 90210, on checkpoints at 20,000 steps.

Read depths follow a six-rung ladder derived per K by one mechanical rule. Because the answer depends only on $h \bmod K$, a ladder chosen without reference to residue structure silently re-asks in-distribution questions at nominally deep h ; every rung’s residue is therefore checked against the training hops $\{1, 2, 3\}$ before the ladder is accepted. The top rung is the smallest h in $[32, 63]$ with $h \equiv K/2 \pmod{K}$: the antipodal residue, the position furthest from any trained hop, evaluated at a squaring count (five) that the rule holds fixed across the entire curve. A second read at a fixed effective distance is carried alongside as a control, so that a change in κ across K can be separated from a change in the depth of the read.

Two training recipes are run at every K : the entity adapter frozen at initialization, or trained alongside the backbone. Both use the same contrastive-plus-cosine auxiliary objective; the adapter is the only axis varied. Three seeds per (K, recipe) .

6.2 PROTOCOL: PRE-REGISTRATION, ADVERSARIAL AUDIT, AND INDEPENDENT RECOMPUTATION

The two chapters share one protocol, described once here.

Bands before data. Every wave’s decision bands, thresholds, and test statistic are written into a dated record before the corresponding cells run, and verdicts are read strictly against those bands. The capability band is CAPABILITY-HOLDS = $\kappa \geq 0.90$ at the antipodal top rung on at least 2 of 3 frozen seeds. The ordering threshold below, $T \geq 53$ of 72, was enumerated together with its exact two-sided p at threshold (0.009868) before any eleven-squaring cell was scored.

Adversarial review before compute. Designs pass independent adversarial rounds whose depth scales with cost: one audit round for eval-only work, an audit plus a separate resource red-team for the breadth wave, and a five-round design gauntlet (draft, attack, revision, verify, revision) for the parameter wave of Section 7, formally closed only when every FATAL-class finding had been discharged. What those rounds caught before launch, stated as a record rather than a claim: two launch-losing defects self-caught by the builder before any audit round (a sequence-length assertion that fails at the two smallest K , and an evaluation index hard-coded to one K ’s hop label that would have raised at the first evaluation of every cell); one further FATAL found independently by both the audit and the red-team (an allow-list bug in the battery scorer); four FATAL-class design defects across the parameter wave’s two adversarial rounds, one of them created by the fix to an earlier one, and one of them a pre-registered target outcome that was arithmetically unreachable at 15 of 16 cells; and, at wave launch, a flat per-step cost ceiling that would have aborted every $K \geq 32$ cell mid-run. Both independent build audits read zero FATAL.

Re-measure clauses. A single-seed reading that exceeds a band is not a finding. The clause, written before the runs, sends it back for re-measurement at an independent evaluation draw (seed 31337); reproduction promotes it to a real reading, non-reproduction records it as an outlier and the verdict stands. Section 6.4’s toehold survived this clause and four other excursions did not.

Mandatory attribution. No scale-degradation claim was permitted to publish before a token-budget attribution arm had run, because the token budget is held fixed across scales and can manufacture a degradation. That clause is what withdrew one of our own results (Section 7.5).

Independent recomputation before publication. Every headline number is recomputed directly from the raw per-cell records by a role separate from the one that wrote the prose. Four rounds of corrections resulted; none changed a verdict, and each is carried on the published note rather than silently absorbed. A control’s floor was overstated (the fixed-distance control reads $\kappa \geq 0.97$ in 35 of 36 cells of the first wave, not all 36; the floor is 0.9470). Two re-measured values had been transcribed from the wrong hop of the right file ($K=20$ at $h=1$ re-measures 0.0664, $K=24$ at $h=52$ re-measures 0.0430; both inside band on either reading). A cell count had double-counted: 42 trained cells plus 6 eval-only anchor re-scores, not 48 trained. A drift-slope extrapolation predicting a wall breach near $K \approx 48$ was retracted outright, because that slope came from a single endpoint segment while global fits over the whole curve read -0.0146 to -0.0162 and support no breach- K prediction at all. In Section 7’s chapter a direction-of-effect tally was corrected from 1/5/6 to 1 helped, 4 flat, 7 hurt under the aggregator’s own ± 0.01 rule, and a wall statement was corrected from a cell-level "all 12" to a reading-level 118 of 120. Compute ledgers were corrected on the same basis: a projected 69 GPU-h became 81.415 measured, and an extension total of 88.424 became a marginal 44.493 once parent spend was no longer re-billed.

The instrument catch. Before either wave, the evaluation harness drew its entity pool from a hard-coded seed while reseeded checkpoints had trained on their own pools. Under that defective instrument the frozen-versus-trainable comparison read as a *complete* separation: all 754 of 754 pairs ordered, $p = 1.34 \times 10^{-10}$. Under matched pools the same checkpoints read 596 of 754 with 216 violating pairs and a median gap of $+0.0098$, a thirty-two-fold shrink. The two trainable quadrants move from 0.7266 to 0.9922 and from 0.2793 to 0.9785 in median recovery, while the two frozen quadrants move by at most 0.0039: frozen adapters are pool-agnostic and trained adapters had specialized to their training entities, which the defective instrument scored as a capability difference. The separation was retracted internally and never published. Every ordering number in Section 6.5 is a matched-pool re-score.

None of this is offered as evidence that the readings below are correct. It is offered as the reason they are reported at the precision they are: each has passed a band fixed before the data existed, an adversarial round before the compute was spent, and a recomputation by someone other than its author.

Table 2: Read-path capability across breadth at 98M parameters, under P1b (exact teacher-forced operator substitution), at each K 's antipodal top rung. κ is chance-corrected retrieval over K slots; $n=256$, eval seed 90210, checkpoints at 20,000 steps; $d = K+1$ throughout. The last two columns give the fixed-distance control's hop and its minimum κ over the six cells at that K . $K=24$ is the eval-only anchor. The curve ends at $K=40$ by construction, not by measurement: the antipodal probe requires $3K/2 \leq 63$ to exist inside the matched five-squaring band, so no reading here licenses any claim about $K \geq 44$ in either direction (Section 6.7). Archives: `experiment-runs/2026-08-22_kscaling_wave0/` (the $K=32$ calibration sextet), `experiment-runs/2026-08-22_kscaling_sweep/` ($K=12$ to $K=32$ and the anchor re-scores), `experiment-runs/2026-08-22_kscaling_frontier/` ($K=36$, $K=40$).

pair (K, d)	top rung h (residue)	κ frozen, 3 seeds	κ trainable, 3 seeds	control h_{fix}	control min κ
(12, 13)	42 (6)	1.0000 / 0.9957 / 1.0000	1.0000 / 1.0000 / 1.0000	40	0.9830
(16, 17)	40 (8)	1.0000 / 1.0000 / 0.9958	0.9958 / 0.9708 / 1.0000	36	0.9958
(20, 21)	50 (10)	1.0000 / 0.9959 / 0.9959	1.0000 / 0.9918 / 0.9959	44	0.9794
(24, 25)	36 (12)	1.0000 / 0.9959 / 1.0000	0.9878 / 0.9878 / 1.0000	52	0.9470
(28, 29)	42 (14)	0.9959 / 1.0000 / 0.9959	0.9959 / 1.0000 / 0.9959	32	0.9878
(32, 33)	48 (16)	0.9960 / 0.9798 / 1.0000	0.9919 / 0.9960 / 0.9919	36	0.9879
(36, 37)	54 (18)	1.0000 / 0.9879 / 0.9960	1.0000 / 0.9839 / 0.9960	40	0.9839
(40, 41)	60 (20)	0.9920 / 0.9960 / 0.9920	0.9880 / 0.9800 / 1.0000	44	0.9720

6.3 READ-PATH CAPABILITY IS FLAT AT CEILING FROM (12,13) TO (40,41)

Forty-eight cells of record carry this curve: 42 freshly trained (six calibration cells at $K=32$, 24 in the main sweep, 12 at the frontier) plus six eval-only re-scores of the $K=24$ anchor onto this ladder.

Every K clears the pre-registered CAPABILITY-HOLDS band on 3 of 3 frozen seeds, not the 2 of 3 the band requires. The floor across all 48 cells is $\kappa = 0.9708$, at $K=16$ in a trainable seed; frozen medians run 0.9920 to 1.0000. The curve does not decline: the two frontier points, added specifically to look for a frontier, read frozen medians 0.9960 at $K=36$ and 0.9920 at $K=40$, with 12 of 12 cells at ceiling. The fixed-distance control reads $\kappa \geq 0.97$ in 47 of 48 cells with a floor of 0.9470 at one anchor cell, so the flatness is not an artifact of the top rung's particular depth.

The calibration gate ran alone before the rest of the curve was licensed. At $K=32$ it moved cross-entropy from 11.037 to 4.528 and read in-distribution $\kappa = 1.000$ at $h \in \{1, 2, 3\}$ with deep $\kappa = 0.9960$ at $h=48$; the license to spend the remaining cells was granted on that measurement rather than assumed from the anchor.

Over a 3.3-fold range of binding load there is no measurable capability slope. Given an exactly written operator, the read path composes it through the antipodal depth at every breadth tested.

6.4 THE LEARNED WRITE IS AT CHANCE AT EVERY BREADTH, WITH ONE TOEHOLD AT $K=12$

The same checkpoints, read through the same path with the model's own learned operator, give the complementary picture. The band is chance ± 3 binomial standard deviations at $n=256$, computed per K ; a breach requires an above-band reading replicated across at least two seeds, and a single-seed excursion goes to the re-measure clause.

The median chance-corrected reading at $h=1$ decays across the curve: 0.0604, 0.0354, 0.0111, 0.0075, 0.0014, -0.0060 , $+0.0016$, -0.0036 at $K=12$ through $K=40$. Across the whole curve 334 of 336 deep-hop readings sit inside band, and both exceptions are single cells that re-measure inside. The verdict is WALL-BREACHED-AT- $K=12$ and WALL-HOLDS at $K \geq 16$.

The $K=12$ breach is reported in full because it is the only one, and its structure is what makes it uninteresting as a capability. The six cells read 0.1211 to 0.1484 at $h=1$, median 0.1387, against chance 0.0833: κ between 0.041 and 0.071, roughly 3.2 binomial standard deviations, about fourteen extra correct answers in 256. It reproduces at an independent evaluation draw, where both re-measured cells read 0.1406, so it is real. It is also confined to $h=1$, the shallowest hop and one

Table 3: The learned-write wall across breadth at 98M parameters, under P0 (the model’s own SGD-learned write, read through the identical path). Band top is chance plus three binomial standard deviations at $n=256$. The fourth column pools the six cells at each K (two recipes, three seeds) at $h=1$, the shallowest and easiest hop; the last column counts readings above band across the six deep-hop rungs in all six cells. Archives as Table 2.

pair (K, d)	chance $1/K$	band top	P0 at $h=1$, min to max (median)	cells above band	deep-hop readings above band
(12, 13)	0.0833	0.1352	0.1211 to 0.1484 (0.1387)	5 of 6	0 of 42
(16, 17)	0.0625	0.1079	0.0703 to 0.1250 (0.0957)	1 of 6	0 of 42
(20, 21)	0.0500	0.0909	0.0430 to 0.1094 (0.0605)	1 of 6	0 of 42
(24, 25)	0.0417	0.0791	0.0273 to 0.0703 (0.0488)	0 of 6	1 of 42
(28, 29)	0.0357	0.0705	0.0273 to 0.0508 (0.0371)	0 of 6	0 of 42
(32, 33)	0.0312	0.0639	0.0156 to 0.0312 (0.0254)	0 of 6	0 of 42
(36, 37)	0.0278	0.0586	0.0156 to 0.0469 (0.0293)	0 of 6	0 of 42
(40, 41)	0.0250	0.0543	0.0117 to 0.0273 (0.0215)	0 of 6	1 of 42

of the three depths the model trains on: at $K=12$, zero of 42 deep-hop readings clear the band. A one-hop advantage at a trained depth that vanishes at every untrained depth is a toehold, not partial competence, and it does not compose.

Four further single-cell excursions occurred and all four failed the re-measure clause: $K=16$ at $h=1$, 0.1250 falling to 0.0977; $K=20$ at $h=1$, 0.1094 falling to 0.0664; $K=24$ at $h=52$, 0.0859 falling to 0.0430; and $K=40$ at $h=4$, 0.0586 against a band top of 0.0543, whose re-measured cell reads 0.0156 at $h=4$ and 0.0391 as its maximum across all ten hops.

Read together with Table 2, the dissociation is between reading and writing, not between shallow and deep. The same weights that compose a supplied operator through the antipodal depth at every breadth cannot produce that operator at any breadth.

6.5 FREEZING BUYS DEPTH ROBUSTNESS, AND THE ADVANTAGE GROWS WITH BREADTH

Whether the entity adapter is frozen or trained turns out not to matter for the capability of Section 6.3 and to matter a great deal for how that capability survives read depth.

The statistic is a stratified within- K exact permutation count. At each K , U_K is the number of the nine frozen-versus-trainable seed pairs in which the frozen cell reads higher, ties counted as one half; $T = \sum_K U_K$ out of nine times the number of strata. Stratifying within K rather than pooling is deliberate: the pooled form mixes strata whose κ ceilings differ, and it was replaced during the design gauntlet for that reason.

At the antipodal top rung, five squarings, the ordering is negligible: $T = 43.0$ of 72 across the eight strata against a threshold of 53 ($T = 32.0$ of 54 across the six strata of the first wave). The verdict is ORDERING-NEGLIGIBLE, and it is what a reader looking only at Table 2 would expect, since both recipes sit at ceiling there.

Extending the read to eleven squarings makes the effect appear.

The eight-stratum statistic reads $T = 61.5$ of 72 against the pre-data threshold of 53, two-sided exact $p = 3.071 \times 10^{-5}$, and the median gap is $+0.0364$. The verdict is ORDERING-ROBUST-CONFIRMED. Dropping any single K leaves the remaining seven strata between 52.5 and 55.5 of 63, which rescales to 60.0 to 63.4 of 72: the threshold is cleared on all eight leave-one-stratum-out subsets. Restricted to the six strata of the wave that preceded the frontier extension, the same eleven-squaring statistic reads 43.5 of 54, reproducing the number that pass had reported.

The result is graded, not a separation, and the paper states it that way. The two frontier strata are perfectly ordered, 18 of 18 pairs at $K=36$ and $K=40$, as are $K=24$ and $K=28$; but four of the eight strata, $K \in \{12, 16, 20, 32\}$, still contain violating pairs. Nor is the statistic monotone in depth: across five, seven, nine, and eleven squarings it reads 48.0, 57.0, 65.0, 61.5, and the dip at eleven traces to a single frozen $K=32$ cell reading $\kappa = 0.8669$, below all three trainable cells in its stratum. That non-monotonicity is reported rather than smoothed.

Table 4: The freeze ordering and the depth cost across breadth, at 98M parameters, under P1b. Columns 2 to 4 give median κ per recipe at eleven squarings and their gap; U_K counts the nine within- K frozen-versus-trainable seed pairs the frozen cell wins (ties one half). The last two columns give the depth cost, the change in retrieval accuracy between five and eleven squarings, per recipe. All readings are matched-pool re-scores (Section 6.2). Archive: experiment-runs/2026-08-22_depthext_across_k/.

K	frozen κ	trainable κ	gap	U_K (of 9)	depth cost, frozen	depth cost, trainable
12	0.9744	0.9744	+0.0000	6.0	−0.0195	−0.0195
16	0.9667	0.9417	+0.0250	6.5	−0.0312	−0.0547
20	0.9465	0.9342	+0.0123	7.0	−0.0469	−0.0625
24	0.9755	0.9307	+0.0448	9.0	−0.0234	−0.0625
28	0.9797	0.9473	+0.0324	9.0	−0.0195	−0.0469
32	0.9637	0.9234	+0.0403	6.0	−0.0312	−0.0742
36	0.9719	0.9036	+0.0683	9.0	−0.0273	−0.0820
40	0.9599	0.8758	+0.0841	9.0	−0.0391	−0.0938

The depth cost carries the breadth interaction. The pre-registered band DRIFT-K-INDEPENDENT compares each K 's drift against the $K=24$ reference; the largest deviation anywhere on the curve is 0.0430, inside the ± 0.05 band, so the band passes at all eight K exactly as pinned. The raw pattern shows something the band was not constructed to detect. The frozen arm is flat with no trend in K , ranging −0.0195 to −0.0469; the trainable arm grows roughly monotonically, from −0.0195 at $K=12$ to −0.0938 at $K=40$. The pinned verdict is therefore true as pinned, and true of the frozen arm outright; for the trainable arm it is a statement about this range of K and nothing more. An earlier attempt to convert that growth into a numeric breach prediction was retracted on independent recomputation (Section 6.2), and no extrapolation past the measured range is claimed here.

The combined reading is one sentence: freezing buys depth robustness that is breadth-stable, and trainable adapters pay a depth cost that grows with breadth. Meanwhile the wall is unmoved by depth: at eleven squarings, 48 of 48 cells sit inside the chance band, the closest approach being a $K=36$ maximum of 0.0547 against a band top of 0.0586.

6.6 COVERAGE AND DEPTH DRIFT

Two supporting measurements at the anchor breadth $K=24$ bound how much of the claim rests on a favorable choice of read.

Coverage. Fifteen previously unmeasured residues were evaluated on both frozen arms at $n=256$, against a pre-registered COMPLETE-COVERAGE band of 0.95 at all fifteen. The floor across those 30 readings is 0.9922; the contrastive-plus-cosine arm reads a minimum of 0.9961 and a mean of 0.9990, the cosine-only arm a minimum of 0.9922 and a mean of 0.9982, and both re-read at a minimum of 0.9961 under matched pools. Together with the three training residues and five previously measured ones, all 23 nontrivial positions of the $K=24$ cycle are now measured at 0.99 or above under exact write, while the model's own learned write reads 0.019 to 0.059 against chance 0.0417 at every one of them. The exact-write ceiling is not an average over a favorable subset of the cycle.

Depth drift. Holding the residue fixed at 13 and varying physical depth over $h \in \{13, 61, 253, 1021, 4093\}$, all congruent to 13 modulo 24, leaves the correct answer identical by construction and changes only the number of squarings, from three to eleven. The pre-registered bands were ROBUST if every deeper reading sits within 0.02 of the $h=13$ reading and DRIFT if the decline is monotone and larger. The verdict is DRIFT: the two frozen arms fall from 1.0000 to 0.9219 and from 1.0000 to 0.9062, strictly monotone across the ladder. Extended across the checkpoint population, 42 of 55 cells are strictly monotone non-increasing and the trainable cosine-only arm is monotone in 8 of 8. Three of the 58 checkpoints predate the configuration flags that define the recipe grid and are reported separately.

The interpretation is narrow and worth stating: repeated squaring is exact in arithmetic and is not exact in floating point, so the composition read degrades slowly with the number of squarings even when the answer it should return does not change. That cost belongs to the read, and the train-

ing recipe modulates it. At eleven squarings the freeze effect is a median gap of +0.1133 while the auxiliary-loss main effect is 0.0137; but within the trainable arm, switching from contrastive-plus-cosine to cosine-only costs -0.2832 , whereas the same switch within the frozen arms moves $+0.0312$ in the other direction. Freezing and the contrastive auxiliary are substitutes for the same robustness, not independent additive gains.

6.7 SCOPE AND LIMITATIONS

The curve ends at $K=40$ for an arithmetic reason, not a measured one. The ladder rule takes the top rung to be the smallest h in $[32, 63]$ congruent to $K/2$ modulo K , and $[32, 63]$ is exactly the band costing five squarings, held fixed across the curve so that breadth is the only moving axis. Such an h exists only when $3K/2 \leq 63$, that is $K \leq 42$. At $K=44$ the derivation raises rather than returning a ladder. An independent audit re-derived both frontier ladders and all six of record from the rule alone, without importing the design’s configuration, and confirmed that the $K=44$ derivation raises, so the limit is a verified property of the construction rather than an assertion. The two ways past it, a different squaring count or a non-antipodal top rung, each break one of the two things the design holds fixed and make the breadth axis uninterpretable; $K=44$ was therefore dropped at design time. No reading on this curve licenses any claim about $K \geq 44$ in either direction, and every table caption carries that limit.

P1b is exact teacher-forced operator substitution, not a learned write. Every capability number in this chapter is a read-path measurement on a supplied operator. The model’s own writes are Section 6.4’s wall, and the two must not be quoted as one result.

The task is a synthetic graft. The composition head and its entity-binding task exist to exercise the read path inside a real language model. Nothing here claims the backbone uses this circuit in ordinary language modeling, and the graft is not a language-modeling benchmark.

Three seeds per (K, recipe) . A per- K rank test on three versus three has a minimum attainable two-sided p of 0.10, so no per- K significance claim is made or is possible; every test above is stratified across K for that reason. This was disclosed in the design, not discovered at harvest.

$d = K+1$ co-varies with K throughout. The manipulated variable is the pair, never K alone, and a curve at fixed d would be a different experiment.

One evaluation seed for the curves of record (90210, $n=256$); the re-measures use 31337. Seed variation here is across training checkpoints, not across evaluation draws.

The ordering result is depth-conditional. It is negligible at the antipodal top rung and confirmed at eleven squarings; it is a claim about a deep read, not about the capability of Section 6.3, and it is graded rather than separating at this scale.

7 PARAMETER SCALING

Section 6 varies binding load at one model size. This chapter holds the task, the ladder, the metric, and the training budget fixed and varies the model instead, from 98M to 392M parameters. The protocol is the one described in Section 6.2.

7.1 THE SECOND SCALE POINT

The reference point is Section 6’s backbone: $d_{\text{model}} = 768$, 12 layers, $d_{\text{state}} = 64$, 97,619,712 backbone parameters, and per-arm totals from 97,816,977 at $K=16$ to 97,860,009 at $K=40$. The second point is $d_{\text{model}} = 1536$, 16 layers, $d_{\text{state}} = 128$, 391,872,512 backbone parameters, per-arm totals from 392,095,889 to 392,175,785: a factor of 4.008 per arm, 4.014 on the backbone alone. Steps (20,000), batch size (32), learning rate (3×10^{-4}), warmup (200 steps), the ladder rule, the metric, the recipes, and the evaluation seed are identical across the two points. Four of Section 6’s eight breadths are ported, $K \in \{16, 24, 32, 40\}$, always with $d = K+1$.

One asymmetry is load-bearing throughout and is stated before any result. The token budget is held fixed rather than scaled with parameters, so tokens per parameter falls from 1.87 to 0.47 at $K=40$. The resulting confound runs in one direction only: an under-trained larger model can manufacture

Table 5: Read-path capability at 392M parameters against the 98M reference, under P1b (exact teacher-forced operator substitution), at each K 's antipodal top rung; κ chance-corrected, $n=256$, eval seed 90210, checkpoints at 20,000 steps, $d = K+1$, three seeds per cell. Both scales train on the identical step, batch, and learning-rate schedule; the token budget is not scaled with parameters (Section 7.1). $K=24$'s 98M cells are the eval-only anchor, which is why the pre-registered sensitivity form drops that stratum. Archives: `experiment-runs/2026-08-22_scaleaxis_sweep/` (392M training and batteries), `experiment-runs/2026-08-22_scaleaxis_stagec/` (depth extension, re-measures, and the 98M reference re-scores), with the 98M originals in `experiment-runs/2026-08-22_kscaling_sweep/` and `experiment-runs/2026-08-22_kscaling_frontier/`.

pair (K, d)	top rung h	κ frozen, 3 seeds (392M)	frozen median (98M)	κ trainable, 3 seeds (392M)	trainable median (98M)
(16, 17)	40	1.0000 / 1.0000 / 1.0000	1.0000	0.9792 / 0.9708 / 0.9917	0.9958
(24, 25)	36	0.9959 / 0.9959 / 0.9959	1.0000	0.9878 / 0.9674 / 0.9837	0.9878
(32, 33)	48	0.9960 / 0.9879 / 0.9839	0.9960	0.9516 / 0.8750 / 0.8065	0.9919
(40, 41)	60	0.9880 / 0.9960 / 0.9880	0.9920	0.7596 / 0.8438 / 0.8798	0.9880

an apparent degradation, but it cannot manufacture a capability. A flat capability curve and a stable wall are therefore safe against it, while every degradation is a candidate artifact until an attribution control rules the budget out. Section 7.5 is that control, and one of the two degradations did not survive it.

The flat shape was pre-registered as the expected null rather than reported as a surprise. The recall-capacity literature already reports structural ceilings that move little with model size: Arora et al. (2023) find associative-recall capacity governed by state dimension rather than parameter count, and Jelassi et al. (2024) find state-limited copying behavior that scale does not lift. The design recorded both as the precedent class for a flat reading before any 392M cell was queued.

7.2 CAPABILITY: NO DETECTABLE DIRECTIONAL SHIFT AT FOUR TIMES THE PARAMETERS

Table 5 gives the per-seed readings at both scale points side by side.

All twelve frozen cells clear the $\kappa \geq 0.90$ gate on 3 of 3 seeds at every ported K . Frozen medians read 1.0000, 0.9959, 0.9879, 0.9880 with a floor of 0.9839, and move between 0.0000 and 0.0081 relative to their 98M counterparts.

The cross-scale test is a stratified exact permutation statistic over eight strata (four K by two recipes) and nine cross-scale seed pairs each, counting how often the 392M cell exceeds its 98M counterpart. It reads $T = 17.0$ of 72, below the degradation bar of 19: nominally a degradation. The pre-registered sensitivity form drops the $K=24$ stratum pair, whose 98M cells are an eval-only anchor trained under an older runner, leaving six strata and $T = 14.5$ of 54 against a degradation bar of 12: inside band. The precedence rule was fixed before data, and says that where the two forms disagree the six-stratum verdict governs. They disagree. The verdict of record is NO DETECTABLE DIRECTIONAL SHIFT at four times the parameters.

The eight-stratum reading is not argued away but reported with what is wrong with it. It is tie-heavy: 14 of its 72 pairs are exact ties, a ceiling effect rather than a signal. It survives only 4 of 8 leave-one-stratum-out subsets, where the six-stratum form is the one the design nominated in advance. And its improvement branch was arithmetically unreachable from the 98M numbers before the wave ran, since the maximum attainable improvement per cell ranged 0.0000 to 0.0122 against a 0.05 threshold, at 0 of 8 cells; a statistic that cannot register the outcome it is nominally testing for is not the one to carry the verdict.

Per K , six of the eight cells are stable within ± 0.05 . The two that are not are both trainable, at $K=32$ (-0.1169) and $K=40$ (-0.1442), and both also fall below the $\kappa \geq 0.90$ gate at 392M. Section 7.5 is about exactly those two cells and about what happened when we tested them.

"No detectable shift" is not "no shift". It is the pre-registered band's label for a reading inside the noise floor at this seed and stratum count, and it should be read as the resolution of the instrument rather than as a proof of invariance.

7.3 THE WALL IS SCALE-STABLE

The learned write was measured at 392M under the same band construction as Section 6.4, chance plus three binomial standard deviations at $n=256$, per K , with the same re-measure clause.

Of the 60 P0 readings at each ported breadth, $K=24$ and $K=32$ read 60 of 60 inside band. $K=16$ reads 59 of 60, the exception a trainable seed at $h=1$ reading 0.1133 against a band top of 0.1079, which re-measures to 0.0859 at the independent draw. $K=40$ reads 59 of 60, the exception a trainable seed at $h=8$ reading 0.0547 against a band top of 0.0543, an excess of a single correct answer in 256, which re-measures to 0.0156 with a cell maximum of 0.0352 across all ten hops. No breach reproduces at any ported K , so the verdict is WALL-HOLDS at all four and, across the two scales, WALL-SCALE-STABLE. Four times the parameters, on this budget, does not teach the model to write the operator.

The extended cells of Section 7.5 give the same reading at 40,000 steps: 118 of 120 P0 readings inside band, with two single-seed $h=1$ excursions ($K=16$ at 0.1133, $K=24$ at 0.0820) unreplicated at their own K and therefore verdict-compatible under the excursion clause. Within 392M alone, the deepest read available puts P0 in band in 24 of 24 cells at fifteen squarings (h from 32,772 to 32,804); that reading is labeled in every record as not comparable to the archived 98M P0 at eleven squarings, and it is not used as a cross-scale claim.

7.4 THE FREEZE ORDERING SHARPENS TO PERFECT SEPARATION

Section 6.5's ordering is graded at 98M: real under a stratified test at depth, with violating pairs in half the strata. At 392M it stops being graded.

The within-scale statistic T_W counts, out of 36, the four ported K times nine frozen-versus-trainable seed pairs the frozen cell wins. At 98M it reads 30.5 at eleven squarings and 25.0 at the antipodal top rung. At 392M it reads 36.0 of 36 at both readouts: the statistic's maximum, with zero realized ties, two-sided exact $p = 1.25 \times 10^{-5}$ under the conditional permutation null. Every frozen seed beats every trainable seed, at every ported breadth, at both readouts. Leave-one-stratum-out reads 27.0 of 27 in all four subsets at both readouts, where the 98M reference clears its own bar of 24 of 27 in only 2 of 4 subsets at eleven squarings (24.0, 21.5, 24.5, 21.5).

The threshold deserves its own sentence, because a weaker one was available. A raw exact $p < 0.01$ at four strata is $T \geq 30$, which the 98M reference itself clears by half a pair; confirming the 392M wave against that bar would have proved nothing about scale. The design instead required $T_W > 31.5$ together with leave-one-stratum-out clearing in at least 3 of 4 subsets, which is to say that confirmation required the 392M wave to be strictly more robust than its own 98M reference. The sub-case of record is ORDERING-SCALE-STRENGTHENS, with descriptive deltas of +5.5 at eleven squarings and +11.0 at the top rung.

This comparison carries no token-budget caveat. Both recipes train on the identical budget at the identical scale, so the confound cancels inside the comparison rather than being argued around it.

Two bounds. Thirty-six of 36 is the ceiling of the statistic, so it cannot read as more than perfect at this resolution, and a larger seed count would be needed to say how much more than perfect it is. And this is a four-stratum test where Section 6.5's was an eight-stratum test; the design compensated by raising the confirmation threshold rather than by treating the shorter instrument as equally strong.

7.5 SCALE BY BREADTH: ONE DEGRADATION WITHDRAWN BY OUR OWN CONTROL, ONE ARCHITECTURAL

The two axes interact, and the interaction is confined to the trainable recipe. At 392M the frozen arm's top-rung κ ranges 0.0121 across the four breadths while the trainable arm's ranges 0.1399; the fixed-distance control shows the same shape, 0.0080 against 0.1040, which is what licenses reading the effect as breadth-driven rather than as a depth cost in disguise.

Read naively, that is two scale degradations, at $K=32$ and $K=40$. The design forbade publishing either one before an attribution arm had run, for the reason given in Section 7.1: the token budget is fixed across scales and can manufacture exactly this shape. Table 6 gives that arm in the order its three controls resolved it.

Table 6: The attribution sequence at 392M, top rung, P1b, trainable recipe. Each control resumes the 20,000-step checkpoints of seeds 0 and 1 for a further 20,000 steps, doubling tokens. V1 tests the token budget at $K=32$; V2 tests it at $K=40$ through the same resume path, which re-opens the schedule’s warmup; V2’ repeats V2 with the learning rate held constant at the schedule’s own 20,000-step floor and nothing else changed. Medians are over the seeds available at that step count (three at 20,000, two at 40,000), and the two-versus-three seed convention gap is disclosed in Section 7.7. Archives: `experiment-runs/2026-08-23_attribution_arm/` (V1, V2) and `experiment-runs/2026-08-23_v2prime/` (V2’), against the 20,000-step parents in `experiment-runs/2026-08-22_scaleaxis_sweep/`.

cell and control	resume schedule	κ at 20,000 steps	κ at 40,000 steps	verdict
$K=32$, V1	warm restart	0.9516 / 0.8750 / 0.8065 (0.8750)	0.9153 / 0.9073 (0.9113)	token-budget-limited
$K=40$, V2	warm restart	0.7596 / 0.8438 / 0.8798 (0.8438)	0.4391 / 0.6635 (0.5513)	control damaged
$K=40$, V2’	constant 3.00×10^{-5}	0.7596 / 0.8438 (parents of the two cells)	0.7276 / 0.8598 (0.7937)	unrecovered and undamaged

$K=32$ is withdrawn. V1 resumed seeds 0 and 1 to 40,000 steps at twice the tokens, and both cleared the $\kappa \geq 0.90$ gate: 0.9153 and 0.9073, median 0.9113, against a 20,000-step median of 0.8750. The below-bar seed rose from 0.8750 to 0.9073 and the above-bar seed settled from 0.9516 to 0.9153. The verdict is TOKEN-BUDGET-LIMITED and the $K=32$ scale-degradation claim is withdrawn. We report this as the protocol working rather than as a result: the control that removed the claim was mandated before the claim existed, and the claim never published.

$K=40$ needed two controls. The first, V2, used the same resume mechanism, which re-opens the linear-warmup-plus-cosine schedule and lifts the learning rate from 3.00×10^{-5} to approximately 1.66×10^{-4} , a 5.5-fold warm restart disclosed before the arm ran. Those cells did not merely fail to recover; they were damaged, reading 0.4391 and 0.6635 against parents of 0.7596 and 0.8438, for seed-matched changes of -0.3205 and -0.1803 , the largest movement anywhere in the arm. A control that damages its own cells cannot rule the budget in or out, so the verdict was recorded as SCALE-DEGRADES, unstrengthened, rather than claiming the pre-registered word on a broken instrument, and the arm was re-run.

The second control, V2’, held the learning rate constant at 3.00×10^{-5} , the schedule’s own 20,000-step floor, for the same two cells over the same marginal 20,000 steps, with the restart removed and nothing else changed. The cells read 0.7276 and 0.8598 against parents of 0.7596 and 0.8438: unrecovered, 0 of 2 above the $\kappa \geq 0.90$ bar, and undamaged, with seed-matched changes of -0.0321 and $+0.0160$ inside the parent band. Doubling the tokens does not recover the degradation and removing the restart does not either. The verdict is SCALE-DEGRADES, cleanly strengthened: at $K=40$ the trainable arm’s degradation at 392M is architectural rather than a budget or schedule artifact. The wall check on those cells holds, with P0 maxima of 0.0391 and 0.0469 against a band top of 0.0543.

The recipe separation therefore widens with scale exactly where breadth is largest. At $K=40$ and 20,000 steps the frozen-minus-trainable gap is 0.144 in κ ; measured against the frozen arm’s own 40,000-step cells (0.9840 and 0.9880) it is 0.192.

The depth tail, in the strong form. One cross-scale degradation survives the attribution arm in full, and it is depth robustness rather than capability. At eleven squarings the cross-scale statistic reads $T = 10.5$ of 72, exact $p = 3.6 \times 10^{-5}$; at the elected depth $s^*=13$ it reads $T = 6.5$ of 72, exact $p = 9.5 \times 10^{-7}$, against a degradation bar of 19. Unlike the capability curve of Section 7.2, the six-stratum sensitivity form agrees with the eight-stratum form here, and leave-one-stratum-out survives 8 of 8. The depth $s^*=13$ was elected by a mechanical rule applied to the 98M re-score before any 392M cell was queued.

The strong form is what the attribution arm established. At twice the tokens all six extended cells move further from recovery rather than toward it; the closest miss is still 0.062 outside the ± 0.095 recovery band; and marginal cross-entropy fell in 10 of 12 extended cells, median -0.125 and largest -2.686 , so the models demonstrably kept training while the tail did not recover. The effect reaches both recipes, not only the trainable one: frozen $K=16$ and $K=24$ read stable (-0.0125 and -0.0285 against 98M), while frozen $K=32$ moves 0.9194 to 0.7863 and then 0.7621 at 40,000

steps, and frozen $K=40$ moves 0.9038 to 0.7837 and then 0.7236. The trainable arm's tail is close to gone at the widest breadth, 0.6715 at 98M against 0.0184 at 392M and 0.0004 after extension.

Two of the twelve extended cells failed a pre-registered clause requiring the marginal segment to reduce cross-entropy, reading +0.026 and +0.030, both at $K=40$. The clause was ruled mis-scoped for a resumed segment already at a plateau, and the flat cross-entropy was recorded as attribution evidence in its own right, since it says the budget was delivered and bought nothing. The scoping decision is on the record rather than applied silently.

7.6 TWO INSTRUMENT FINDINGS

Two results here are about the measurement apparatus rather than about the model, and both are reported because the second one changed a verdict.

Warm restarts destabilize trained trainable adapters at the breadth frontier, and nothing else measured here. Under the identical restart, every other extended cell stays within ± 0.04 at the top rung: $K=16$ trainable at -0.0083 and $+0.0208$, $K=24$ trainable at -0.0041 and -0.0041 , $K=32$ trainable at -0.0363 and $+0.0323$, $K=32$ frozen at $+0.0040$ and $+0.0040$, $K=40$ frozen at -0.0040 and -0.0080 . Only the $K=40$ trainable cells move by -0.3205 and -0.1803 . A trained trainable adapter at the widest breadth is the fragile object, and frozen adapters shrug the restart off. The practical consequence for anyone extending a fast-weight adapter from a checkpoint is that the resume should either hold the learning rate constant or disclose the schedule it re-opens, because the schedule alone can move a frontier cell by a third of the metric's range.

Direction of effect. Extension arms are ordinarily licensed on an unstated assumption that more training cannot hurt. Scoring all twelve (cell by readout) comparisons of 20,000-step against 40,000-step medians against a ± 0.01 flat band gives 1 helped, 4 flat, and 7 hurt. The single case that helped is the $K=32$ top rung at $+0.0363$, the cell that withdrew our own claim. One of the seven that hurt is a boundary call, $K=24$ trainable at the top rung at -0.0102 , a thousandth past the band, and it is named as such rather than counted silently; the honest restatement is that the extension helped once and hurt or was flat everywhere else. Checking that assumption rather than asserting it is what forced V2' and converted the $K=40$ verdict from standing on a broken control to standing on a clean one.

7.7 SCOPE AND LIMITATIONS

Two model sizes are not a scaling law. 98M and 392M are two points. No slope is fitted, "survives four times the parameters" is exactly and only what was measured, and nothing here extrapolates to a billion parameters or to a different architecture family.

The token budget is fixed and its confound is one-directional. Tokens per parameter falls from 1.87 to 0.47 at $K=40$. Under-training can manufacture a degradation and cannot manufacture a capability, so the flat capability curve of Section 7.2 and the stable wall of Section 7.3 are safe against it, while every degradation required the attribution arm. One of the two degradations did not survive that arm. Nothing in Section 7.5 should be cited as evidence that the capability is scale-fragile.

The extended arm is two seeds, not three. The 40,000-step cells are seeds 0 and 1, compared against 20,000-step medians over three seeds. A seed-matched sensitivity that restricts the 98M twin to the same two seeds leaves 5 of 6 depth-tail verdicts unchanged and flips only frozen $K=32$, the cell nearest the bar. The convention gap is recorded as a design gap of this chapter, not resolved by it.

The $K=40$ trainable depth-tail row at 40,000 steps comes from the schedule-damaged cells. V2' was scored on the breadth battery and not on the depth ladder, so that one row is a warm-restart reading. The verdict does not turn on it: the 20,000-step reading already missed the ± 0.095 recovery band by 0.558.

"Double the tokens" means tokens seen, not loss moved. At $K=40$ in particular the marginal cross-entropy barely moved, +0.026 and -0.055 under the restart and -0.048 and $+0.005$ at constant learning rate. Those cells are at a plateau, so the $K=40$ token-budget explanation is ruled out on budget delivered rather than on further fitting achieved.

V2' is a two-cell schedule control at one breadth and one constant learning rate. It rules out this restart as the cause of this damage. It is not a learning-rate sweep, and it does not establish that no schedule could recover the $K=40$ trainable cells.

Four breadths, not eight. $K \in \{12, 20, 28, 36\}$ have no 392M counterpart, and $K=44$ is construction-impossible for the arithmetic reason given in Section 6.7. Three seeds per $(K, \text{recipe}, \text{scale})$, with every test stratified across K for the reason given there. The task remains the synthetic graft of Section 6.1, and nothing here claims the backbone uses this circuit in ordinary language modeling.

Two recorded process disclosures. The chapter's compute ledger overran its pre-registered 130 GPU-h gate at approximately 135.7 GPU-h all-in; the overrun was licensed by a recorded adjudication before the cells were queued, and is reported as authorized rather than fitted after the fact. And the fixed-distance control's three pre-registered labels were defined on κ without naming an arm, so the two arms return different labels; the control is reported per arm rather than elected, and the ambiguity is recorded as a design gap.

8 RELATED WORK

Associative capacity of fast-weight states. Nichani et al. (2024) prove that under argmax decoding a rank-one matrix can recover on the order of d associations, which is why no result in this paper derives a rank or capacity claim from argmax recall: the rank law of Section 3 forces exact continuous recovery through a Procrustes-aligned cosine readout, and every argmax recall number in Section 4 carries their caveat explicitly. Their analysis is the reason the causal razor of Section 3 refuses argmax decoding anywhere a rank claim depends on it.

Associative recall in transformers. Multi-query associative recall is a regime where softmax attention is documented as strong: Arora et al. (2023) introduce the MQAR benchmark and show attention solves it where sub-quadratic models struggle, Jelassi et al. (2024) report the same advantage for copying over state-space models, and even two-layer from-scratch transformers reliably develop induction circuitry (Olsson et al., 2022). Our transformer baseline's chance-level reading therefore runs against the expectation this literature sets, and we do not interpret it as an architectural inability. The leading candidate explanation was optimization: the arm trained at the shared default learning rate with no architecture-specific search on the recall task, and its training loss is near flat across the run (Section 4.1). A subsequent four-point learning-rate search on the recall task itself rules this out: the best-optimizing rate reads recall furthest below chance, so a better-optimized language-modeling objective does not surface recall on this arm (Section 7 reports the full grid). The separation verdict accordingly rests on the vector-state ablation comparison alone, with the transformer now disclosed as a second failing baseline whose chance-level reading survives an explicit learning-rate search; Section 7 reports the completed measurement and its optimization-recall dissociation.

Rank measurement in linear attention. Two contemporaneous studies (Nazari & Rusch, 2026; Sun et al., 2026) measure the rank of linear-attention states descriptively, on frozen checkpoints. Section 3 differs in both directions of inference: the force-rank arms intervene at train time, and the necessity/sufficiency step at $d_{\min}$ is a causal result a descriptive measurement cannot license.

Key normalization and stability. The qk-normalization line (Kimi Linear, Team et al., 2025; Qwen3-Next, Qwen Team, 2025) conditions single-vector eigenvalue stability of the state update. The attractor of Section 5 is a different object, cross-key population geometry. It was measured with qk normalization active throughout, and removing the normalization is a within-noise null at $n=3$.

DeltaNet and gated variants. DeltaNet (Yang et al., 2025b) and Gated DeltaNet (Yang et al., 2025a) supply the architecture substrate for the capability and pathology legs (Sections 4 and 5) and benchmark it for quality and throughput. This paper asks what the states represent: the rank law of Section 3 is established on a separate matrix-state encoder family with no delta-rule write, while the causal recall capability and the write-induced geometry at scale are established on the delta-rule substrate itself. The gating axis reappears here only as a measured factor in Section 5.2, where it trends toward amplifying the attractor without confirming.

Prior negative result in this program. A published companion study ("The Gradient Does Not See Rank," ICML 2026 MI workshop; Larson, 2026) showed that bolting a matrix state onto a pretrained

backbone through a flatten-then-project readout leaves the gradient rank-blind. The present paper is the from-scratch counterpart: when the state is native and the readout preserves structure, SGD does see rank (Section 3), and the resulting medium supports a capability its matched ablations lack (Section 4).

9 DISCUSSION AND LIMITATIONS

Scale scope of the capability legs. The rank law and the recall separation are small-model, synthetic-task results by design: the encoder family trains in minutes and the head-to-head contender is a 14M-parameter model on a controlled binding task. The strength of that regime is causal closure (train-time rank forcing, state zeroing, frozen margins); its limit is external validity. Nothing here shows the recall separation surviving at language-model scale or on natural data, and the matched-budget caveat bounds every transformer comparison: a differently scaled or differently trained transformer requires its own matched re-run before any extrapolation. The learning-rate search for the transformer arm on the recall task itself has since run. Under the identical frozen protocol, a grid of four learning rates spanning 10^{-4} to 3×10^{-3} (three seeds each, 20,000 matched steps) reads at or below chance at every point: mean acc_A is 0.0142 at $\text{lr} = 10^{-4}$, 0.0283 at 3×10^{-4} (the shared default used elsewhere in this paper), 0.0320 at 10^{-3} (the best of the grid, $1.02\times$ chance), and 0.0283 at 3×10^{-3} ; 0 of 12 individual (learning rate, seed) cells and 0 of 4 grid means clear the frozen 0.09375 demonstration bar ($3\times$ chance at $K = 32$). Per the pre-registered decision rule, the transformer now joins the parameter-matched vector-state ablation as a second baseline that fails to demonstrate recall after an explicit learning-rate search; the ablation comparison remains the registered verdict carrier (Section 4.2), and this measurement changes no registered tier.

The grid also closes the leading unexcluded explanation named in Section 6 for the chance-level reading: the best-optimizing learning rate (10^{-4} ; final training loss 6.55 to 6.60, well below the shared default’s approximately 7.5) reads $\text{acc}_A = 0.0142$, 6.3 standard deviations below chance, while the two worse-optimizing learning rates (10^{-3} and 3×10^{-3} ; final loss approximately 7.7 to 7.8) read closer to chance. Optimization quality and recall dissociate on this arm: giving the transformer the learning rate that best optimizes the language-modeling objective does not surface recall, so the chance-level reading is not attributable to an under-optimized transformer. This closes the specific unexcluded explanation named above; it does not establish an architectural impossibility. The matched-budget caveat still bounds the reading: this is a two-layer, approximately 15M-parameter, 20,000-step, from-scratch regime, and the MQAR and induction-head literature’s positive transformer results (Section 6) live at other scales, architectures, and budgets.

One stress point is a locate-only null. At the disclosed above-capacity stress load ($K/d = 0.75$), the completed three-arm table reads chance in every arm: contender 0.0189, ablation 0.0195, transformer 0.0218, against chance 0.0208. The table is locate-only by pre-registration, not a verdict-grade cell; no capability claim is made at this load, and it bounds the single-hop recall separation to loads below this stress point at this training budget.

Scope of the pathology leg. The ladder result is a geometry claim: span fraction worsens monotonically with scale, but this paper does not tie the drift to a downstream performance cost, and validation loss is neutral in every mitigation cell measured. The tie between the geometric drift and any behavioral cost is the outstanding scientific question, and the 392M token-budget confound (Section 5.3) caps what the transfer wave can say across scales.

Multi-hop composition is open. The two-hop task is verdict-INDETERMINATE, with partial recall in three of nine pooled contender seeds, none generalizing to held-out hop depths (Section 4.5), and a separate compositional-depth program in this research line is at the design-and-calibration stage with a pre-registered lens question resolved but no model verdict; we claim neither. Depth generalization is future work, stated as such.

Instrument validity as a recurring risk. Three times in this program (Sections 2.5, 3.3, 4.3), the first instrument pointed at a true phenomenon read false. In each case the fix was minor once the defect was seen. Representational claims about matrix states need independent checks of the kind Section 2.5 catalogs, because errors in either direction arise from instrument choice alone at this scale: the wrong-tap probe read a real capability as absent, and the identity-padded target sold ambient structure as recovery.

10 CONCLUSION

A $d \times d$ state written by outer products is a measurable representational medium. Its trained rank is set causally by the task’s representation theory in the encoder family. In the delta-rule family, the first layer’s state causally carries recall that a matched vector-state ablation lacks (a compute-matched transformer also fails, after an explicit learning-rate search); the write keys of its language models drift toward collapse at every larger scale, stock normalizer or no. The storage and the drift come from one mechanism, and that is the design constraint this paper leaves behind: interventions aimed at the geometry must operate on the write rule itself rather than on its normalizers, with the Section 5 negative transfer result as the first measured bound on that search. A recall-task learning-rate search for the transformer baseline has since run (Section 7): the transformer fails at every point in the grid, strengthening the capability separation to a two-baseline result without moving the registered verdict, which the vector-state ablation still carries. One measurement would extend the map from here: a tie from the geometric drift to a behavioral cost.

REFERENCES

- Simran Arora, Sabri Eyuboglu, Aman Timalsina, Isys Johnson, Michael Poli, James Zou, Atri Rudra, and Christopher Ré. Zoology: Measuring and improving recall in efficient language models, 2023. URL <https://arxiv.org/abs/2312.04927>.
- Samy Jelassi, David Brandfonbrener, Sham M. Kakade, and Eran Malach. Repeat after me: Transformers are better than state space models at copying, 2024. URL <https://arxiv.org/abs/2402.01032>.
- Samuel Larson. The gradient does not see rank: Rank-indifference in matrix-CODI on prosQA. In *Mechanistic Interpretability Workshop at ICML 2026*, 2026. URL <https://openreview.net/forum?id=Spof4PusVI>.
- Philipp Nazari and T. Konstantin Rusch. The key to state reduction in linear attention: A rank-based perspective, 2026. URL <https://arxiv.org/abs/2602.04852>.
- Eshaan Nichani, Jason D. Lee, and Alberto Bietti. Understanding factual recall in transformers via associative memories, 2024. URL <https://arxiv.org/abs/2412.06538>.
- Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. In-context learning and induction heads, 2022. URL <https://arxiv.org/abs/2209.11895>.
- Qwen Team. Qwen3-Next-80B-A3B-Instruct, 2025. URL <https://huggingface.co/Qwen/Qwen3-Next-80B-A3B-Instruct>. Model release and card, Hugging Face, September 9, 2025.
- Ao Sun, Hongtao Zhang, Heng Zhou, Yixuan Ma, Yiran Qin, Tongrui Su, Yan Liu, Zhanyu Ma, Jun Xu, Jiuchong Gao, Jinghua Hao, and Renqing He. State rank dynamics in linear attention llms, 2026. URL <https://arxiv.org/abs/2602.02195>.
- Kimi Team, Yu Zhang, Zongyu Lin, Xingcheng Yao, Jiaxi Hu, Fanqing Meng, Chengyin Liu, Xin Men, Songlin Yang, Zhiyuan Li, Wentao Li, Enzhe Lu, Weizhou Liu, Yanru Chen, Weixin Xu, Longhui Yu, Yejie Wang, Yu Fan, Longguang Zhong, Enming Yuan, Dehao Zhang, Yizhi Zhang, T. Y. Liu, Haiming Wang, Shengjun Fang, Weiran He, Shaowei Liu, Yiwei Li, Jianlin Su, Jiezhong Qiu, Bo Pang, Junjie Yan, Zhejun Jiang, Weixiao Huang, Bohong Yin, Jiacheng You, Chu Wei, Zhengtao Wang, Chao Hong, Yutian Chen, Guanduo Chen, Yucheng Wang, Huabin Zheng, Feng Wang, Yibo Liu, Mengnan Dong, Zheng Zhang, Siyuan Pan, Wenhao Wu, Yuhao Wu, Longyu Guan, Jiawen Tao, Guohong Fu, Xinran Xu, Yuzhi Wang, Guokun Lai, Yuxin Wu, Xinyu Zhou, Zhilin Yang, and Yulun Du. Kimi linear: An expressive, efficient attention architecture, 2025. URL <https://arxiv.org/abs/2510.26692>.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks, 2024. URL <https://arxiv.org/abs/2309.17453>.

Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving mamba2 with delta rule, 2025a. URL <https://arxiv.org/abs/2412.06464>.

Songlin Yang, Bailin Wang, Yu Zhang, Yikang Shen, and Yoon Kim. Parallelizing linear transformers with the delta rule over sequence length, 2025b. URL <https://arxiv.org/abs/2406.06484>.

A THE c^*I COMPLEMENT SCAFFOLD

A mechanism-level observation, architecture-conditional and offered as a candidate rather than a claim of the main thesis. In the converged matrix-state encoder family (the Section 3 substrate’s sibling trained on parallel key-value binding), the trained state decomposes as $Z \approx c^*I_d + \Delta$, where Δ is a rank- $(K-1)$ task correction: the whole-state law holds at 0.5 to 2.9 percent Frobenius residual, with per-example identity alignment $\tau \geq 0.9997$. The effective rank of $Z - c^*I$ sits within 0.3 of the $K-1$ target at the $K=8$ load (deviations 0.03 to 0.26); at the $K=12$ load the three converged runs read 10.22 to 10.41 against the target of 11, deviations of 0.59 to 0.78. The rank- $(K-1)$ characterization is therefore exact-order at the lighter load and approximate at the heavier one. The scaffold is emergent: the architecture has no identity path or output gain that parameterizes it.

The scaffold earns its keep as a loss-blind health signal: the Procrustes residual of the complement correlates with held-out task cosine at Spearman $\rho = -0.973$ across all eleven archived runs, read entirely from a subspace the loss never constrains, so it can grade convergence without labels. Its boundary is architectural, because the scaffold does not exist where Sections 4 and 5 live: in delta-rule states the orthogonal complement channel is numerically empty (complement energy fraction at most 3.2×10^{-12} across 12 of 12 production-family runs), because outer-product writes confine the state to the span of observed keys and values by construction. The scaffold is therefore an encoder-family instrument and a delta-rule impossibility, which is itself a datum: the two families that share the rank law of Section 3 and the capability of Section 4 do not share their complement structure, and mechanism stories that depend on an identity scaffold cannot port across that boundary.

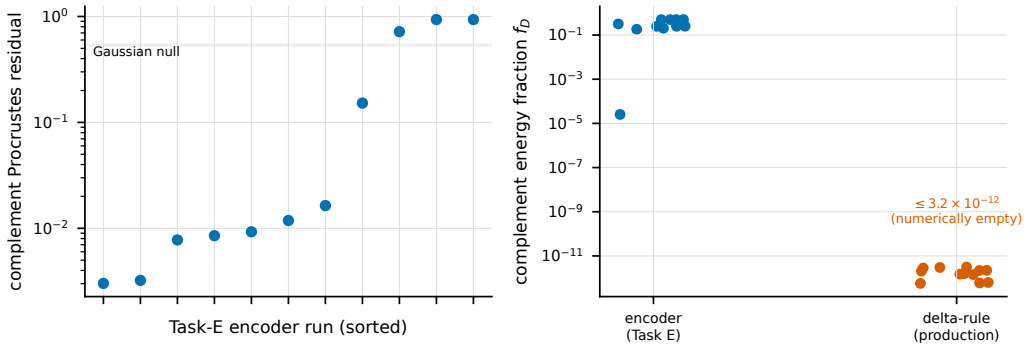


Figure A1: The complement scaffold is architecture- conditional. Left: complement Procrustes residual per archived Task-E encoder run (sorted; log scale). Converged runs sit at 0.003 to 0.018, two orders of magnitude below the Gaussian-null band (0.53 to 0.54, shaded); the high-residual runs are the archive’s non-converged members, which is the separation behind the loss-blind health signal (Spearman $\rho = -0.973$ against held-out task cosine, all eleven runs). Right: complement energy fraction f_D by architecture family (log scale): the encoder family carries measurable complement energy, while delta-rule production states read at most 3.2×10^{-12} , numerically empty, because outer-product writes confine the state to the span of observed keys and values.

B CONFIGURATIONS, FULL TABLES, AND REPRODUCIBILITY

B.1 MODEL AND TRAINING CONFIGURATIONS

Setting	Value
Stage-1 encoder (Section 3)	$h=32$, 3 layers, $d_{state} = d_{min} + 2$ per group; single-state bottleneck; step budgets 8K/20K/20K/8K/40K for $S_3/S_4/A_5/S_5/A_6$
Head-to-head arms (Section 4)	contender $d_{model}=256$, 2 blocks, $d_{state}=64$, 14,049,408 params; ablation 14,048,384 params; transformer depth 2, FLOP-matched $\leq 5\%$; all arms 20,000 steps, matched tokens, seeds identical across arms
Ladder rungs (Section 5)	14M/98M/392M/1.31B = $(d_{model}, L, d_{state})$ of (256, 2, 64) / (768, 12, 64) / (1536, 16, 128) / (2560, 22, 128); measured 14,048,896 / 97,618,176 / 391,869,440 / 1,311,135,488 params
Span-fraction anchors	random $\sqrt{K(K-1)/d}$, collapse $\sqrt{K(K-1)}$; at $K=64$: 7.94/63.50 ($d=64$), 5.61/63.50 ($d=128$)

B.2 PER-SEED AND PER-CELL TABLES

The full per-seed and per-cell tables live in the checksummed archive named in Section B.3, at the archive root linked there, rather than in this appendix: the 19-cell unconstrained rank table with band edges (R1), the 39-cell first-sweep force-rank diagnostic with each cell’s deviation from the $\sqrt{k/d_{state}}$ ceiling, the three cells outside 0.07 named (R1b), the 30-cell force-rank grid plus the 24-cell S_3 extension (R3), the 27-cell head-to-head sweep with all re-metric readings and hard-stop legs (R4), the localization and tap tables for both arms (R5), the per-rung ladder trajectories (R6), the 12-cell 2-by-2 factorial (R7), the 28-cell fix-at-scale wave with both instruments and all gate cells (R8), and the 90-pass memory-matched fan-out (R10). Table 1 reproduces the per-seed R4 verdict values in the main text; every other reading above loads from the named artifacts under the checksums of Section B.3 and the archive’s per-run manifests.

B.3 ARTIFACT MANIFEST AND FIGURE REGENERATION

Every figure in this paper is generated by a single versioned script (`figures/figure-gen.py`) that loads only archived result files and asserts the MD5 checksum of each file before plotting; a checksum mismatch fails the build rather than plotting stale data. The manifest below lists the verdict-bearing artifacts (checksums recorded 2026-07-09/10; the archive layout carries per-run manifests verified against the compute host).

Result	Artifact	MD5
R1/R2	2026-07-09_capability_sweep_harvest/harvest_summary.json	7dce77dcba724cd1004419ac71fe5f2f
R3	2026-07-09_m3fix_harvest/harvest_analysis_output.txt	77be9c3b092c70e83ff08a0261575815
R3 (ext)	2026-07-09_m3fix_s3ext/harvest_analysis_output.txt	d4413385f9f45b71ff9707354ed6b055
R4	2026-07-10_h2h_sweep_harvest/ (27 training + 18 re-metric JSONs; margins token 58fe68e7b15728739a5176d7e591e204)	per-file manifest
R5	2026-07-09_h2h_tap_localization/results/tap_localization_{contender,ablation}.json	362333c8..., ff8e352a...
R6	2026-07-06_trackc_rung3/probe_analysis_rung3.json; 2026-07-06_trajectory_probes/trajectories_tidy.json	6a627c31..., 0fe53d8b...
R7	2026-07-09_attrrob_2x2_escalation_harvest/{box_results/AGGREGATE.json, n3_recompute_summary.json}	d7e4d6b4..., b4bdfdf...
R8	2026-07-10_fixscale_harvest/fixscale_harvest_verdict.json	f2f0aae8...
R9	2026-07-09_zdump_complement/complement_results.json	03208eda...
R10	2026-07-10_h2h_mstar/MSTAR_VERDICT.json	4f115ad5...

Full 32-character checksums for the abbreviated entries appear in the manifest released alongside the figure script (`figures/ARTIFACT_MANIFEST.txt`). The archive root is <https://github.com/saml212/learned-representations/tree/main/experiment-runs>; the anonymized build links an anonymized mirror in its place.

B.4 PRE-REGISTRATION RECORDS

Each verdict traces to a design-document record written before the confirmatory run: the rank-law sweep and its instrument repair (§1.33–§1.36a of the capability-separation design), the head-to-head margins freeze and sweep verdict (§1.31–§1.41 of the head-to-head design), the ladder and mitigation waves (the scale-transfer design and §13 of the frozen-bias design). The records fix scope, seeds, bars, and tiers; where a record and an artifact ever disagreed, the raw-artifact reading and the recorded tiebreak stand in place of the record’s prose.